

TRBOnet

Digital Technologies



Market: Utilities
Company: Loesk

System Specifications:

- 8 re-transmitters
- More than 300 radio units – Motorola DP/DM3401/3601/4401/4601
- Radio System Control Server at every location
- Dispatch console TRBOnet Enterprise and Friendly control units
- TRBOnet Swift IP connectors for base radio stations

Benefits:

- Expansion of radio coverage for dispatch centers by integrating disparate radio networks to one per location via IP connectors
- Motorola IP Site Connect Solution
- Flexible architecture offered by TRBOnet Enterprise
- Monitoring of geolocation of users and movement of vehicles
- Increased quality of the radio signal and voice communications in difficult topographic and electrical disturbance conditions
- Availability of guaranteed reserve radio communication capacity in case of emergency and force majeure situations
- Possible integration with telemetry sensors and systems
- Possible operational control of personnel on station by facility management right from their desks
- Ongoing recording and journaling of communications, data and location in case of emergency situations

Products:

TRBOnet Enterprise
TRBOnet Swift Agent



SUMMARY

Loesk is one of the largest electric infrastructure companies in the energy sector in the Leningrad area and supplies reliable and uninterrupted electricity to 25 cities, large municipalities and industrial areas in the Leningrad area (which has a population of over 1 million). The company has 15 locations which service more than 7,800km of electricity transmission infrastructure (from 0.4 to 110kW) as well as 2400 transformer substations. The company's infrastructure supports more than 2 billion kW/hr of capacity transmitted. The drastic increase in recent year of both economic and social development of the Leningrad area very quickly highlighted the need for more capacity and more reliability in electric transmission. Reliable and protected radio communications is one of the most important and indeed, required conditions of effective and safe operations dealing with electric energy generation and transmission.

SOLUTION

The specialists of Neocom Software were challenged with the following requirements: provide uninterrupted dispatch and control of the communications system in both normal conditions as well as extreme or emergency situations, tie together wide-ranging areas of company locations together into a single network while being limited to a small range of frequencies as well as provide the capability of monitoring various transport types and personnel movement within the 15 company locations.

The most communication-challenged areas were chosen as a pilot of the solution's function. These areas either had sub-standard levels of communications in place or were completely without coverage. The testing of the pilot infrastructure was completely successful and confirmed the superiority of digital radio signaling over that of analog and the client decided on a strategy of implementing this solution at all of their locations.

The first stage of work included specialists from Neocom installing conventional professional-grade DMR radio communications systems in the dispatch centers at Viborg, Kingisepskovo and Slanzevo locations. The next stage deployed the solution to Polprozhsch, Lodeinopolks, Volchovsk, Boksitogorsk and Kirovsk areas of Leningrad. Finally, the last deployments went to Luga, Kirishi, Tichvin and Sosnovi Bor.

When selecting digital client terminals, the recommended units were Motorola Solutions DP/DM/4401/4601 with GPS capability which can locate each individual active unit's location.

To increase reliability of the solution there was a recommendation made to use IP connector software called TRBOnet Swift, which connects base stations to the radio server via IP networks. The reason for the selection of this enhancement was the compact size of the solution as well as the greater reliability, automation (reduction of human error), as well as the compatibility with a specific unit console (Friendly).

Dispatch software TRBOnet Enterprise established single systems of communications in the company locations. Additional desirable functionality was made available such as the monitoring, recording and review of all communications, data as well as geolocation of radio unit users. Further integration steps of the newly implemented digital radio infrastructure will comprise of combination of the individual company location control and dispatch systems to a single system, thereby enabling the creation of a single centralized dispatch center.

RESULTS/BENEFITS

Digital radio systems enabled dispatchers and registered network users establish directed selective radio calls as well as group communications where required. Text communication became possible as well as the GPS monitoring of users and vehicles. Lastly it has enabled the use and centralization of built-in telemetry sensors as well as the recording, reviewing and monitoring of all ongoing conversations, data and telemetry recordings.

The implementation of the solution proposed by Neocom Software has increased the controllability of the system, increased quality of both processes but also quality and clarity of speech over the radio network especially where distortions would typically garble transmission. Additionally, the selection of the DMR standard for Loesk has significantly increased the capacity of the radio system and therefore has increased financial efficiency overall.

Sales and Support

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